



Digimon Ontology Development Report

DOISR Group 3

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TABLE OF CONTENTS

Introduction	2
Development	2
Ontological Requirements	2
Reused Ontological Resources	3
Reused Non-Ontological Resources and Terminologies	3
Conceptual Model Design	4
Ontology Implementation in OWL	5
Main Design Decisions	6
Ontology Evaluation and Improvement Process	6
Documentation, Publication and FAIRness Assessment	7
Knowledge Graph Creation	8
SHACL Validation	8
Additional Tools Used	8
Conclusions	9
Ontology Mapping and SHACL Adaptation	10
References	11

Introduction

The objective of this project is the development of an ontology for the Digimon domain as part of the DOISR course at Universidad Politécnica de Madrid.

The ontology was designed to semantically represent Digimon entities, their stages, attributes, abilities, combat moves and digivolution relationships. The project also includes the creation of RDF knowledge graphs, SHACL validation shapes and FAIR publication mechanisms following Semantic Web and Linked Data principles.

The ontology was implemented using Protégé and Chowlk, documented with Widoco and evaluated using OOPS! and FOOPS!. Additionally, SHACL constraints were defined to validate the generated knowledge graph instances.

This report describes the ontological requirements, design decisions, reused resources, evaluation process, FAIRness assessment and publication mechanisms developed throughout the project.

Development

Ontological Requirements

The ontological requirements were defined during the initial phase of the project based on the analysis of the selected Digimon datasets and the main use cases to be supported by the ontology. Their objective was to determine the scope of the ontology, identify the relevant concepts and define the relationships and information to be represented semantically.

A total of 28 functional requirements were specified using the ontology requirements template provided during the course. These requirements were mainly derived from datasets such as DigiDB_digimonlist.csv, DigiDB_movelist.csv, DigiDB_supportlist.csv, Skills.csv, Skills by Digimon.csv, Digivolutions.csv and Digivolution Requirements.csv.

The requirements focused on representing Digimon entities, their statistics, attributes, moves, skills, support skills and digivolution relationships. They were also aligned with the main use cases of the project: *Battle Team Planning* and *Digivolution Path Analysis*.

During this process, several modelling decisions were made regarding how dataset fields should be represented. Numerical values such as HP, SP, power or memory were modeled



as datatype properties, while concepts such as Digimon, Move, Skill or Digivolution were represented as ontology classes.

Overall, the ontological requirements established the foundation for the conceptual model, OWL implementation, knowledge graph generation and SHACL validation developed in the following stages of the project.

Reused Ontological Resources

As part of the ontology development process, different existing Semantic Web resources and ontologies were explored in order to determine whether any ontology related to the Digimon domain could be reused. The objective of this task was to identify reusable vocabularies or ontology components that could reduce modelling effort and improve interoperability.

However, no complete ontology specifically focused on the Digimon domain was found. Most of the available resources consisted of fan databases, tabular datasets or non-ontological resources, which were useful as data sources but not suitable for direct ontology reuse.

For this reason, the core domain model was developed specifically for this project. Concepts such as Digimon, Move, Skill, Support Skill, Attribute, Stage and Digivolution were modelled according to the information available in the selected datasets and the requirements defined during the initial phases of the project.

Although no domain-specific ontology was reused, several standard Semantic Web vocabularies and technologies were incorporated into the implementation. RDF, RDFS and OWL were used for the ontology representation and modelling process, while XSD datatypes were used to represent numerical and textual values.

Additionally, metadata vocabularies such as Dublin Core Terms (DCTerms) and VANN were reused in order to describe ontology metadata including title, creators, contributors, version information, license and namespace declarations.

This approach allowed the ontology to remain adapted to the Digimon domain while still following common Semantic Web and interoperability practices.

Reused Non-Ontological Resources and Terminologies

Several non-ontological resources were used as the main source of information for building the Digimon ontology. These resources mainly consisted of structured CSV datasets

containing information about Digimon entities, statistics, moves, skills, support skills and digivolution relationships.

The selected datasets were part of the Digimon Cyber Sleuth Dataset and the DigiDB dataset, including files such as *DigiDB_digimonlist.csv*, *DigiDB_movelist.csv*, *DigiDB_supportlist.csv*, *Skills.csv*, *Skills by Digimon.csv*, *Digivolutions.csv* and *Digivolution Requirements.csv*.

These resources were considered non-ontological because they were originally represented as tabular data rather than in Semantic Web formats such as RDF or OWL. However, their structure made them suitable for transformation into an ontology-based model, since dataset fields and relationships could be mapped to ontology classes, object properties and datatype properties.

The datasets were used to define the main ontology concepts, including Digimon, Move, Skill, Support Skill and Digivolution, as well as their associated relationships and properties. In particular, the digivolution datasets were fundamental for modelling source and target Digimon together with the conditions required for each digivolution process.

The selected datasets were aligned with the scope and use cases of the project, especially *Battle Team Planning* and *Digivolution Path Analysis*. During the modelling process, numerical values such as HP, SP, attack, defense, intelligence, speed, power and SP cost were represented as datatype properties, while domain concepts were represented as ontology classes.

Overall, these non-ontological resources provided the basis for transforming structured tabular information into a semantic representation based on OWL, RDF and Linked Data principles.

Conceptual Model Design

The conceptual model of the ontology was developed based on the ontological requirements and the analysis of the selected Digimon datasets. Its objective was to provide a high-level semantic representation of the main entities, relationships and attributes within the Digimon domain before the implementation phase in OWL.

The conceptualization process was carried out using the Chowlk notation, which allows ontology elements such as classes, object properties and datatype properties to be visually represented in a structured and understandable way. The resulting conceptual model included more than 20 ontology elements, including ontology classes, object properties and datatype properties.



The central concept of the ontology is the class *Digimon*, which represents digital creatures together with their associated characteristics and relationships. Several subclasses were defined to classify Digimon according to their type, including *VirusDigimon*, *VaccineDigimon*, *DataDigimon* and *FreeDigimon*. Additional classes such as *Move*, *Skill*, *SupportSkill*, *Attribute*, *Stage*, *Digivolution* and *DigivolutionRequirement* were also incorporated into the model.

The conceptual model also defines several object properties representing relationships between ontology entities. Examples include the relationships between Digimon and their moves, skills, stages, attributes and support skills, as well as the relationships involved in digivolution processes. In particular, digivolutions were modeled as n-ary relations through the class *Digivolution*, which connects source Digimon, target Digimon and digivolution requirements.

Datatype properties were used to represent numerical and literal information associated with Digimon entities. These properties include statistics such as HP, SP, attack, defense, intelligence, speed and memory. Such values were modeled as datatype properties because they represent quantitative characteristics rather than independent domain concepts.

The resulting conceptual model provided the foundation for the OWL implementation of the ontology and ensured consistency between the datasets, the identified requirements and the semantic representation developed throughout the project.

Ontology Implementation in OWL

After the conceptualization phase, the ontology was implemented in OWL using Protégé and Chowlk. OWL was selected as the ontology language because it provides a formal and machine-readable representation of semantic knowledge while supporting reasoning and interoperability within the Semantic Web.

The implementation process involved the creation of ontology classes, object properties, datatype properties and individuals according to the conceptual model previously defined. Classes such as *Digimon*, *Move*, *Skill*, *SupportSkill*, *Attribute*, *Stage*, *Digivolution* and *DigivolutionRequirement* were implemented together with their corresponding relationships and restrictions.

Object properties were used to represent semantic relationships between entities, including associations between Digimon and their moves, skills, attributes, stages and digivolution paths. Datatype properties were implemented to represent numerical and literal information such as HP, SP, attack, defense, intelligence, speed and memory values.



Several metadata annotations were also incorporated into the ontology implementation. These included information related to ontology title, creators, contributors, version, license and publication details using vocabularies such as Dublin Core Terms and VANN.

The ontology was serialized in multiple RDF formats, including Turtle, RDF/XML, N-Triples and JSON-LD, in order to improve interoperability and facilitate publication and reuse. The final OWL implementation constituted the basis for the generation of the knowledge graph, SHACL validation and FAIR publication mechanisms developed in later stages of the project.

Main Design Decisions

Several design decisions were made to keep the ontology consistent and aligned with the project requirements. The class Digimon was defined as the central concept of the ontology, since the selected datasets mainly describe Digimon entities, their statistics, attributes, abilities and digivolution relationships.

Digimon categories — Virus, Vaccine, Data and Free — were modeled as subclasses of Digimon to support classification and querying. A distinction was also established between domain concepts and simple values: concepts such as Digimon, Move, Skill, SupportSkill, Attribute, Stage, Digivolution and DigivolutionRequirement were represented as classes, while numerical information such as HP, SP, attack, defense, speed or memory was represented through datatype properties.

Digivolution was modeled as an independent class in order to represent source Digimon, target Digimon and evolution requirements together. Move, Skill and SupportSkill were also kept as separate classes to reflect the distinctions present in the datasets.

Object properties were used to represent relationships between ontology entities, while controlled individuals such as Fire, Water, Light or Dark were introduced to avoid inconsistencies in attribute values. Finally, metadata vocabularies such as Dublin Core Terms and VANN were reused to describe ontology metadata including title, creators, license and version information.

Overall, these decisions resulted in a semantic model capable of representing the Digimon domain in a clear, consistent and reusable way.

Ontology Evaluation and Improvement Process

The ontology was evaluated using OOPS! and FOOPS!, following an iterative improvement process. The initial OOPS! evaluation identified several issues related to missing

annotations, incomplete domain and range definitions, and undeclared inverse relationships.

To address these problems, labels, comments and additional metadata were incorporated into the ontology, and domains and ranges were defined for the main properties. A second OOPS! evaluation confirmed that most of the initial issues had been resolved. The remaining warnings mainly concerned missing disjointness and inverse relationships, which were considered acceptable for the current scope of the project.

The ontology was also evaluated with FOOPS! to assess its FAIRness level. The first evaluation obtained a score of 72%, mainly due to insufficient metadata and documentation. To improve this result, additional metadata such as creators, contributors, license, provenance, version information and namespace declarations were added. The ontology was also published through a persistent w3id.org URI with content negotiation support.

After these improvements, the FOOPS! score increased to 81%, confirming that the ontology achieved a better level of documentation, accessibility and FAIR compliance.

Documentation, Publication and FAIRness Assessment

The ontology was documented using Widoco, which automatically generated human-readable HTML documentation from the OWL ontology. The generated documentation includes information about ontology classes, object properties, datatype properties, individuals, metadata and namespace declarations, improving the readability and accessibility of the ontology.

The ontology was published online through GitHub Pages and linked to a persistent URI using w3id.org. Content negotiation mechanisms were configured to allow access to both HTML documentation and RDF serializations, including Turtle, RDF/XML, N-Triples and JSON-LD formats.

The ontology was released under the Creative Commons Attribution 4.0 (CC BY 4.0) license in order to support reuse and distribution. Metadata describing the ontology, including title, creators, contributors, version information, namespace declarations and provenance details, were incorporated using Dublin Core Terms and VANN vocabularies.

FAIRness was evaluated using FOOPS!, which assessed aspects such as metadata completeness, persistent identifiers, RDF accessibility, documentation quality and ontology availability. After improving metadata and publication mechanisms, the ontology achieved a FOOPS! score of 81%, indicating a good level of compliance with FAIR principles.

Knowledge Graph Creation

A sample RDF knowledge graph was created in Turtle format (data.ttl) to represent Digimon entities according to the developed ontology. The graph includes examples of Digimon, stages, attributes, moves, skills, support skills and digivolution relationships using the ontology classes and properties.

Representative instances such as Agumon and Greymon were defined together with their statistics, attributes, stages and abilities. Digivolution examples were also included through instances of the *Digivolution* and *DigivolutionRequirement* classes.

Additionally, several intentionally invalid examples were incorporated in order to test the SHACL validation process. These invalid instances included missing values, incorrect datatypes, invalid relationships and negative numerical values.

The knowledge graph therefore served both as an illustrative RDF dataset and as a validation dataset for testing the SHACL constraints.

SHACL Validation

SHACL shapes were defined in the file *shapes.ttl* to validate the RDF knowledge graph according to the ontology requirements. The validation was carried out using SHACL Playground, and the resulting report was stored in *reportShacl.ttl*.

The shapes included constraints related to cardinality, datatypes, numerical ranges, class membership and digivolution structures. Additional logical and structural constraints such as closed shapes, disjointness constraints and inverse path constraints were also incorporated.

The validation process produced both positive and negative results, as expected, since the knowledge graph intentionally included invalid examples to test the behaviour of the shapes. The detected errors included missing properties, invalid datatypes, incorrect class assignments, negative values and invalid digivolution relationships.

Overall, the validation results confirmed that the SHACL shapes correctly enforced the ontology constraints and that the generated knowledge graph was consistent with the ontology specification.

Additional Tools Used

Several additional tools were used throughout the development process in order to support ontology modelling, validation, publication and collaborative work.



Protégé was used for ontology editing and OWL implementation, while Chowlk was employed to design the conceptual model using a graphical notation. Widoco was used to automatically generate ontology documentation, and OOPS! and FOOPS! were used for ontology quality and FAIRness evaluation.

GitHub and GitHub Pages were used for version control and ontology publication, while w3id.org was employed to provide persistent URIs and content negotiation support. SHACL validation was carried out using RDF validation tools compatible with Turtle serializations.

Additionally, generative AI tools such as ChatGPT Plus were used as support resources during different stages of the project, primarily to assist in improving the clarity, structure and consistency of the ontology documentation, conceptualization and SHACL validation process.

The tool was also used to support the refinement of ontology requirements, the organization of ontology elements, the drafting of documentation sections, the interpretation of evaluation results and the generation of example data and validation constraints.

All ontology design decisions, modelling choices, implementation details and evaluation processes were reviewed and validated by the authors. The AI tool was used exclusively as a complementary aid to improve consistency and quality, ensuring that the final ontology and documentation reflect the authors' understanding and work throughout the project.

Conclusions

This project involved the development of a complete ontology for the Digimon domain following Semantic Web and ontology engineering principles. The ontology was designed to represent Digimon entities, stages, attributes, moves, skills, support skills and digivolution relationships using OWL and RDF technologies.

Throughout the project, several ontology engineering tasks were carried out, including requirements analysis, conceptualization, OWL implementation, knowledge graph generation, SHACL validation, ontology evaluation and FAIR publication. Tools such as Protégé, Chowlk, Widoco, OOPS! and FOOPS! were used to support the different stages of development.

The evaluation and improvement process contributed to enhancing the quality, documentation and FAIRness of the ontology. In addition, the publication through GitHub



Pages and w3id.org enabled persistent access and content negotiation for both HTML and RDF representations.

Overall, the project provided practical experience in ontology modelling, Semantic Web technologies and Linked Data publication. The resulting ontology constitutes a reusable semantic resource for representing knowledge related to the Digimon domain and can be extended in future versions with additional concepts, constraints and interoperability mechanisms.

Ontology Mapping and SHACL Adaptation

An interoperability analysis was carried out between the Digimon ontology and the ontology developed by the assigned group (Group 1), which focused on the Pokémon domain.

Both ontologies share a similar modelling approach based on OWL and RDF technologies. In both cases, the central concept is a creature class (Digimon / Pokemon) connected to other ontology entities through object properties and datatype properties. Numerical statistics such as HP, attack, defense or speed were represented as datatype properties in both ontologies, while semantic concepts such as types, abilities or classifications were modeled as ontology classes.

Several equivalent concepts were identified between the two ontologies. For example: the Digimon ontology class Attribute is conceptually similar to the Pokémon ontology class Type, while classes such as Skill, SupportSkill or Move correspond to concepts related to Pokémon abilities and attacks. Both ontologies also make use of metadata vocabularies such as RDF, RDFS, OWL, XSD, Dublin Core Terms and VANN.

However, important structural differences were also identified. The Digimon ontology explicitly models digivolution processes through the classes Digivolution and DigivolutionRequirement, whereas the Pokémon ontology does not include an equivalent structure for evolution requirements. Similarly, the Pokémon ontology defines additional concepts such as regions, habitats, egg groups or body shapes, which are not represented in the Digimon ontology.

As a consequence, several modifications would be necessary in order to adapt the SHACL shapes developed for the Digimon ontology to the Pokémon dataset. These changes would mainly involve replacing namespaces, updating class and property references, and adapting constraints according to the structure of the target ontology. Digimon-specific constraints related to digivolution processes would also need to be removed or redesigned to match the concepts available in the Pokémon ontology.



Overall, the analysis showed that both ontologies follow similar Semantic Web modelling principles despite representing different fictional domains. The comparison also highlighted the importance of ontology interoperability and the need to adapt SHACL validation rules when applying them across different ontology schemas.

References

1. Digimon Database: <https://www.kaggle.com/datasets/rtatman/digidb>
2. Digimon Cyber Sleuth Dataset:
<https://www.kaggle.com/datasets/lianebrisebois/digimon-cyber-sleuth-dataset>
3. Github Repository of Group 1 from DOISR:
<https://github.com/JunhaoZhg/Developing-Open-Interoperable-Semantic-Resources>
4. All the PDFs used in DOISR course this year and tools/webs explained in class (Protegé, OOPS!, FOOPS!, Open Refine, W3ID, WIDOCO, etc.)